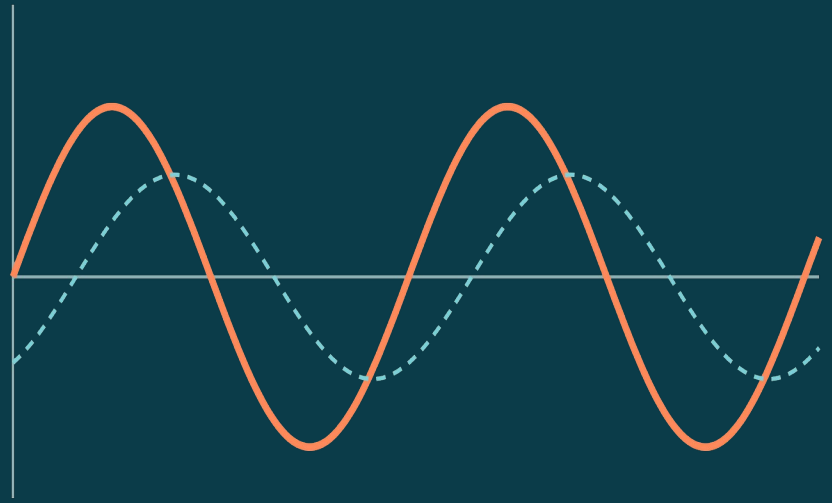
R, L, C Series Circuits and Power



Basic Electrical and Electronics Engineering

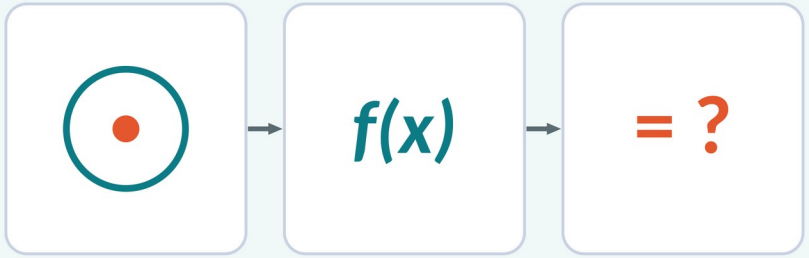
Unit 1: Circuit Fundamentals

Topic: R, L, C, RL, RC, RLC series circuits; real, reactive, apparent power; power factor

Prof. Mitesh Astik

Learning Outcomes

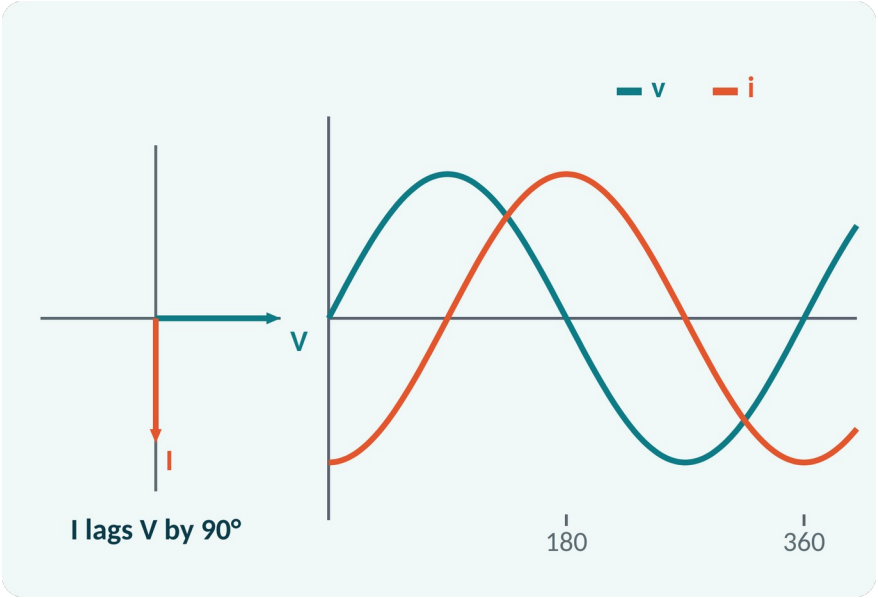
- Find current and phase angle in series R, L, C circuits.
- Draw impedance triangles and phasor diagrams.
- Calculate P, Q, S and power factor.
- Explain why power factor matters.



Linked to CO1
Analyse DC and AC electrical circuits to determine circuit parameters.

Recap: What You Already Know

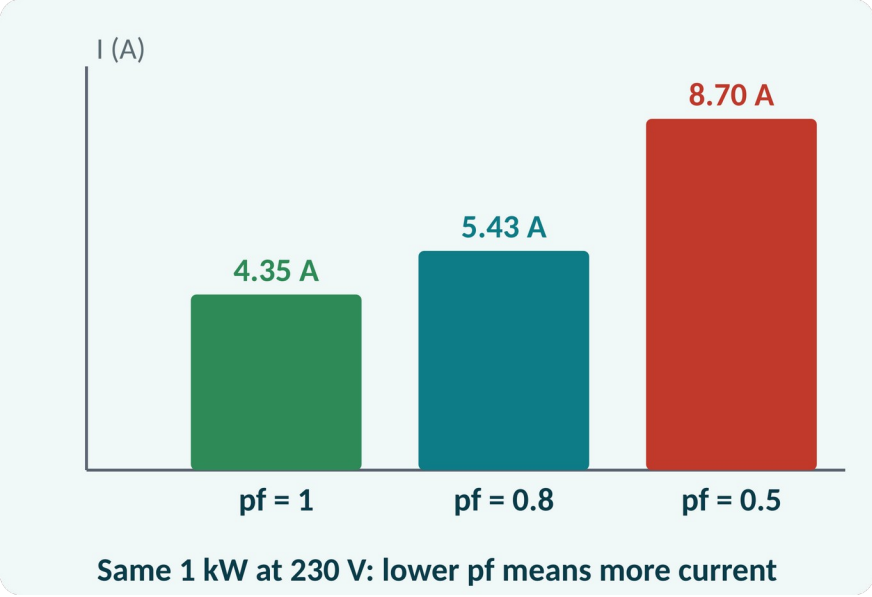
- Phasor V = V∠φ with RMS magnitude.
- j rotates a phasor by +90°.
- Add phasors in rectangular form.
- ω = 2πf; 50 Hz mains.



V = Vm/√2; j = √−1 = 1∠90°
Phasor: RMS length and phase angle

Motivation: Why Power Factor?

- Motors and tube lights are inductive loads.
- Reactive current heats lines but does no work.
- Utilities penalise low power factor.
- Capacitors in the panel correct it.

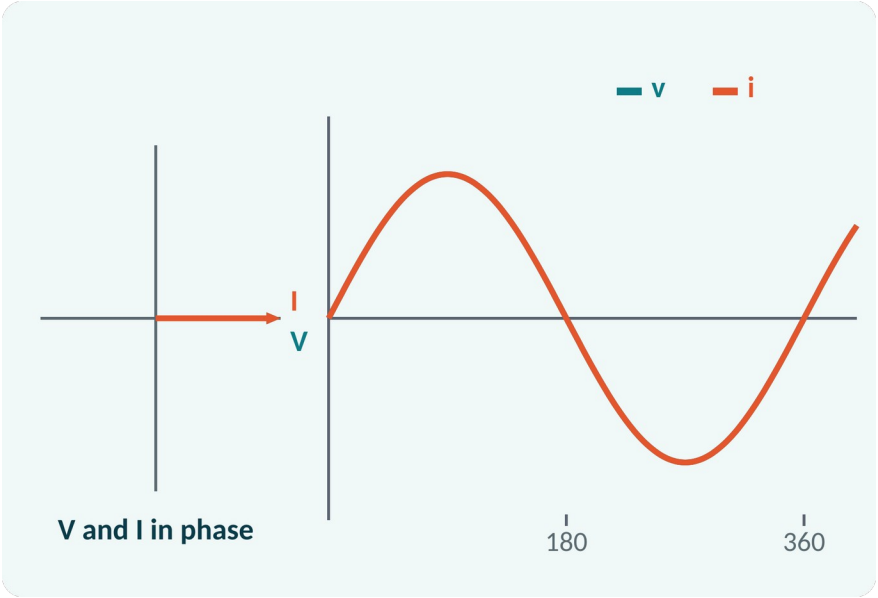


Same 1 kW at 230 V: lower pf means more current

For the same real power, lower pf means more current and bigger losses

Pure Resistor

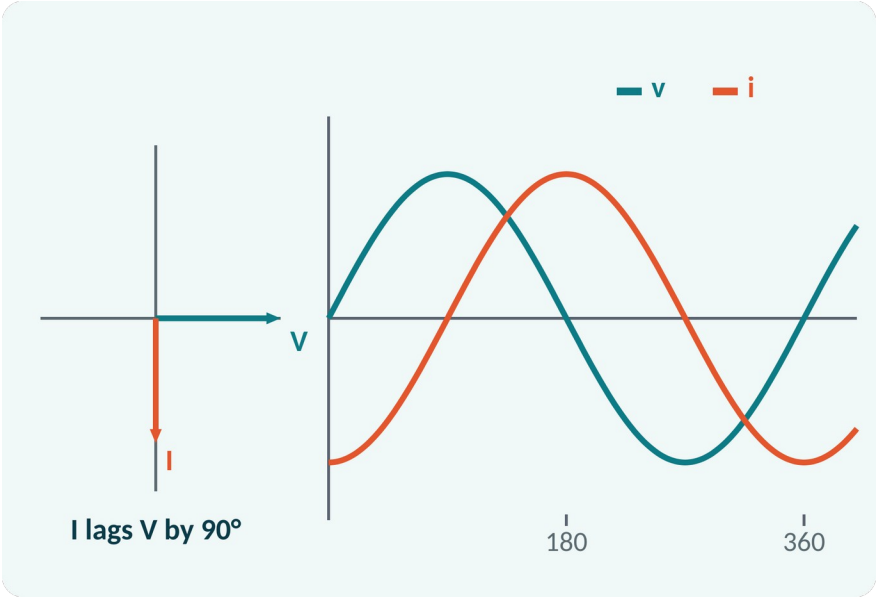
- v = Vm sin ωt gives i = (Vm/R) sin ωt.
- Voltage and current are in phase.
- Impedance equals R, independent of frequency.
- Average power P = VI = I^2R.



V = I·R, φ = 0°, P = V·I = I^2R

Pure Inductor

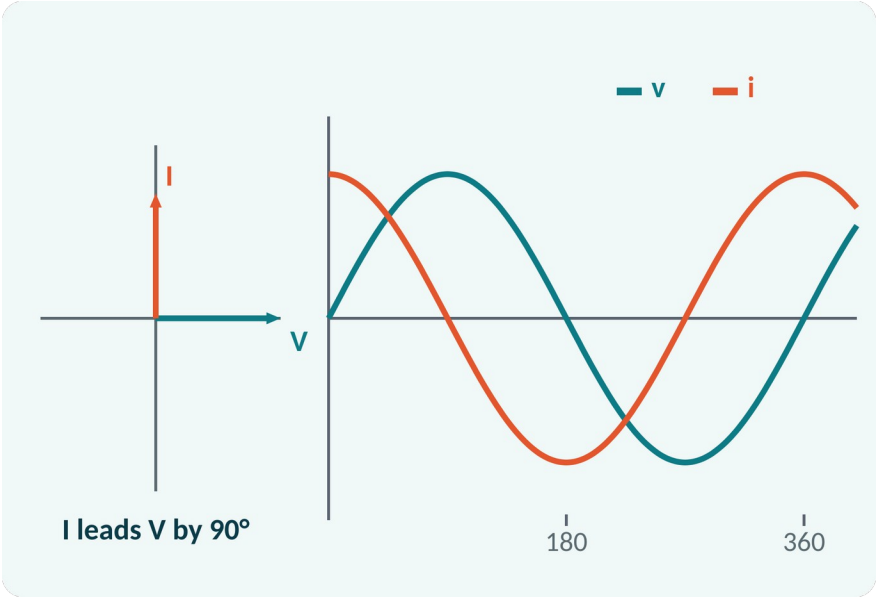
- v = L·di/dt, so current lags voltage.
- Current lags by 90°.
- Inductive reactance XL = ωL = 2πfL.
- Average power is zero; energy is stored and returned.



XL = 2πfL (Ω); I = V/XL; φ = −90°
L henry, f hertz

Pure Capacitor

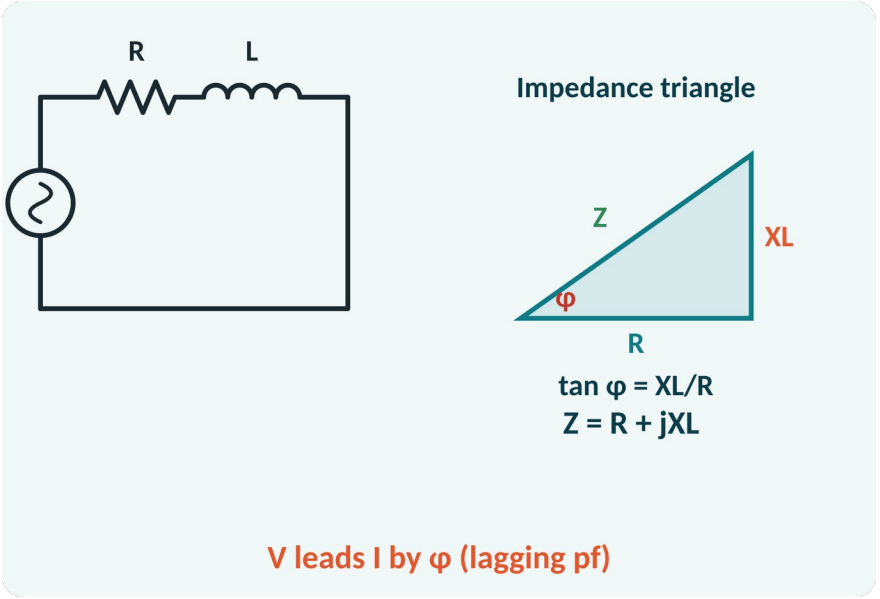
- i = C·dv/dt, so current leads voltage.
- Current leads by 90°.
- Capacitive reactance XC = 1/(2πfC).
- Average power is zero.



XC = 1/(2πfC) (Ω); I = V/XC; φ = +90°
C farad, f hertz

Series RL Circuit

- Same current flows through R and L.
- VR in phase with I; VL leads I by 90°.
- Impedance Z = R + jXL.
- Current lags the supply voltage by φ.

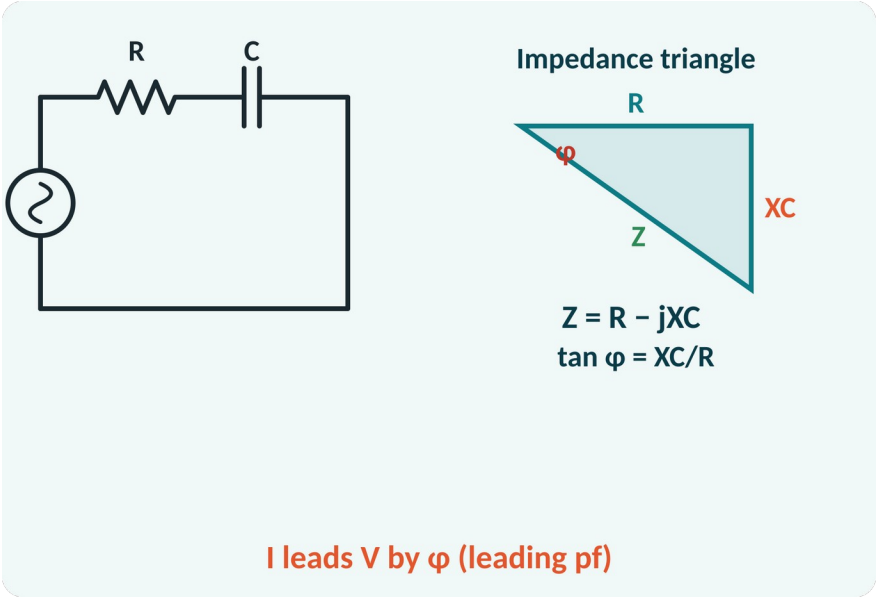


$|Z| = \sqrt{(R^2 + XL^2)}$; $\varphi = \tan^{-1}(XL/R)$; $I = V/|Z|$

$V^2 = VR^2 + VL^2$

Series RC Circuit

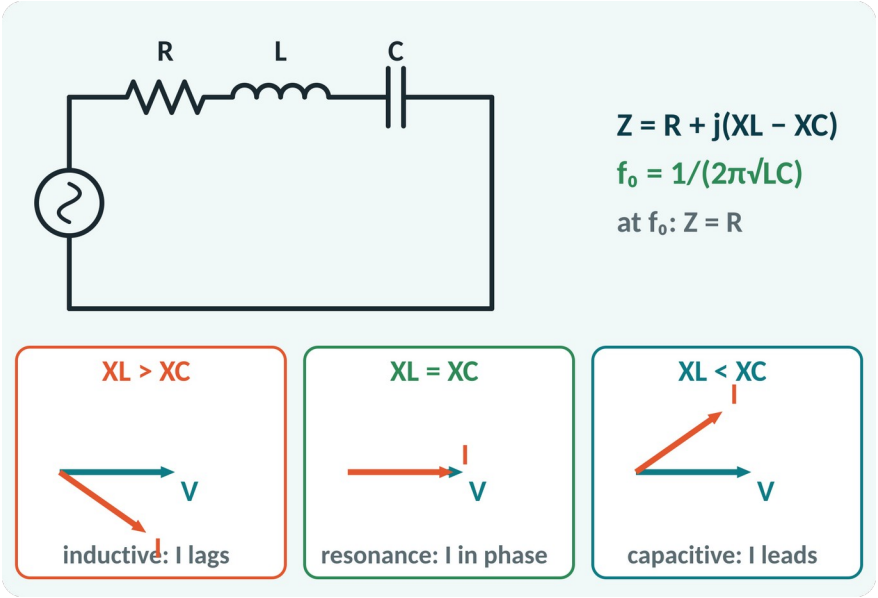
- VR in phase with I; VC lags I by 90°.
- Impedance Z = R − jXC.
- Current leads the supply voltage by φ.
- Used in lag compensators and filters.



$$|Z| = \sqrt{(R^2 + XC^2)}; \quad \varphi = \tan^{-1}(XC/R); \quad I = V/|Z|$$

Series RLC Circuit

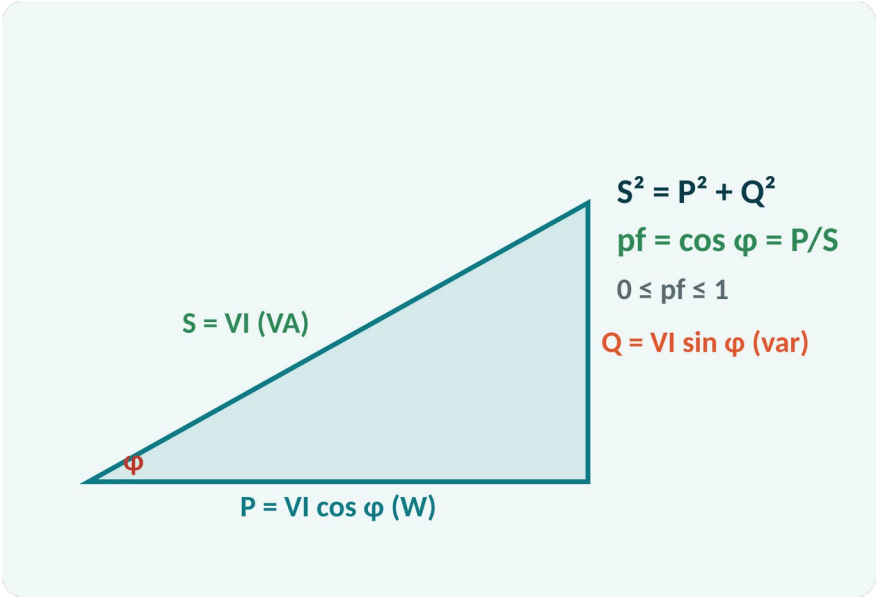
- $Z = R + j(X_L - X_C)$.
- $X_L > X_C$: inductive, current lags.
- $X_L < X_C$: capacitive, current leads.
- $X_L = X_C$: resonance, Z = R, I maximum.



$$|Z| = \sqrt{[R^2 + (X_L - X_C)^2]}; \quad \varphi = \tan^{-1}[(X_L - X_C)/R]; \quad f_0 = 1/(2\pi\sqrt{LC})$$

Real, Reactive and Apparent Power

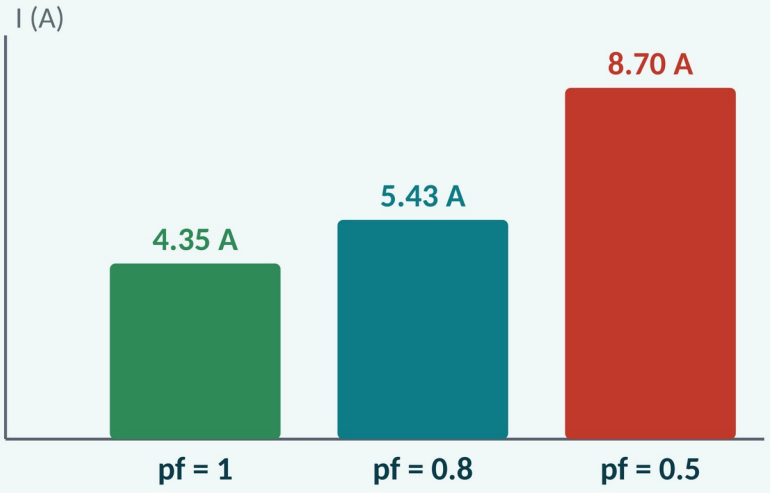
- Real power P is dissipated in R.
- Reactive power Q is exchanged with L and C.
- Apparent power S = V·I is the rating of equipment.
- Power triangle: P, Q, S.



$P = VI\cos\varphi = I^2R$ (W); $Q = VI\sin\varphi = I^2X$ (var); $S = VI$ (VA)
$S^2 = P^2 + Q^2$

Power Factor

- pf = cos φ = P/S.
- Lagging for RL loads, leading for RC loads.
- Low pf: more current for the same P.
- Correction: add a capacitor in parallel.
- Utility tariffs reward pf near 1.

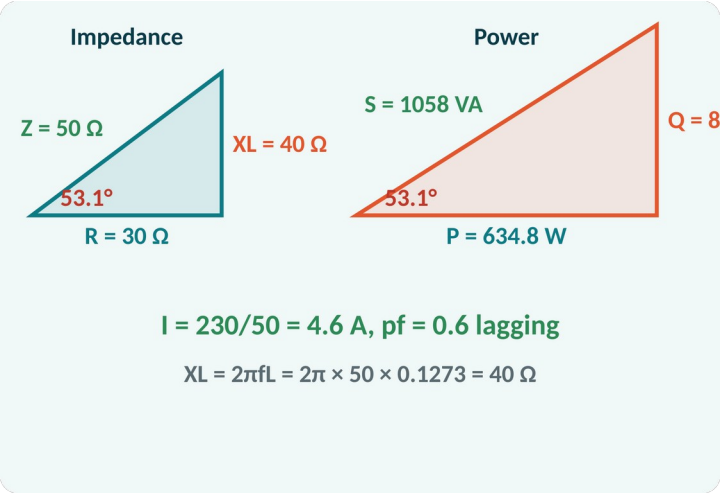


Same 1 kW at 230 V: lower pf means more current

pf = cos φ = R/|Z|; I = P/(V·pf)

Solved Example 1: Series RL

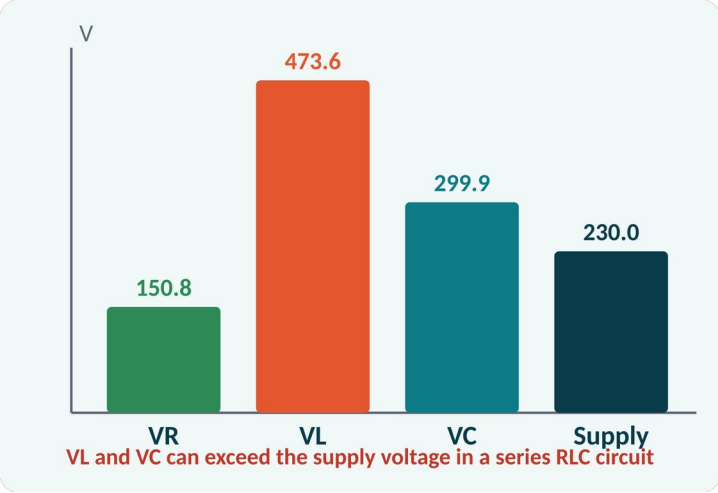
- R = 30 Ω in series with L = 0.1273 H on 230 V, 50 Hz. Find Z, I, φ, pf, P, Q, S.
- 1. $X_L = 2\pi \cdot 50 \cdot 0.1273 = 40\ \Omega$.
- 2. $Z = 30 + j40 \rightarrow |Z| = 50\ \Omega$.
- 3. $I = 230/50 = 4.6$ A; $\varphi = \tan^{-1}(40/30) = 53.1°$.
- 4. pf = cos 53.1° = 0.6 lagging.
- 5. $P = I^2R = 634.8$ W; $Q = I^2X_L = 846.4$ var.
- 6. $S = VI = 1058$ VA.



Answer: I = 4.6 A lagging by 53.1°, pf = 0.6, P = 635 W, Q = 846 var, S = 1058 VA

Solved Example 2: Series RLC

- R = 20 Ω, L = 0.2 H, C = 80 μF on 230 V, 50 Hz. Find I, voltages, pf, P and f_0.
- 1. $X_L = 2\pi \cdot 50 \cdot 0.2 = 62.83\ \Omega$; $X_C = 1/(2\pi \cdot 50 \cdot 80 \times 10^{-6}) = 39.79\ \Omega$.
- 2. $Z = 20 + j23.04 \rightarrow |Z| = 30.5\ \Omega$; $I = 7.54$ A.
- 3. $\varphi = \tan^{-1}(23.04/20) = 49.0°$, pf = 0.656 lagging.
- 4. $V_R = 150.8$ V, $V_L = 473.6$ V, $V_C = 299.9$ V.
- 5. $P = I^2R = 1137$ W.
- 6. $f_0 = 1/(2\pi\sqrt{LC}) = 39.8$ Hz.



Answer: I = 7.54 A, pf = 0.656 lag, P = 1137 W, f_0 = 39.8 Hz

Common Mistakes and Exam Tips

Common mistakes

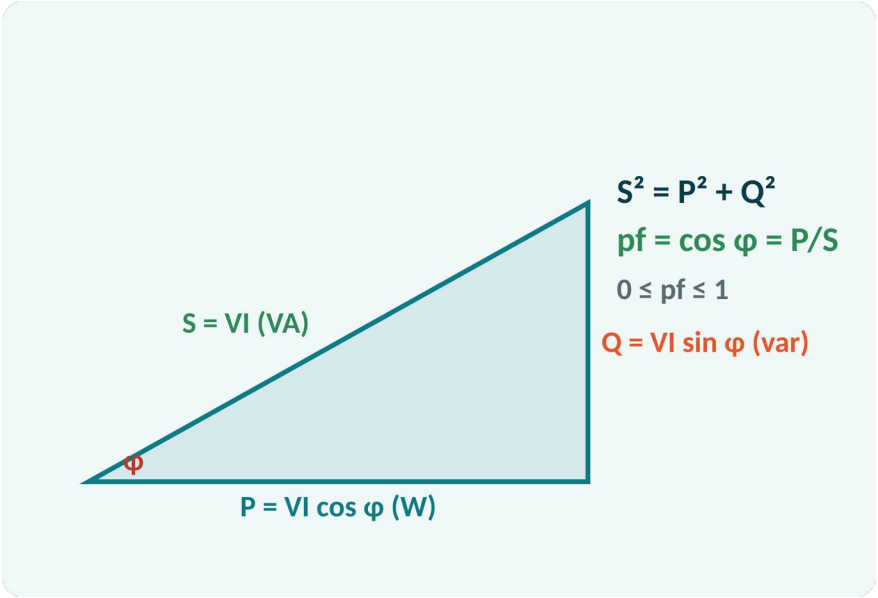
- Adding VR, VL and VC arithmetically.
- Using XC = 2πfC.
- Taking pf = sin φ.
- Forgetting lead or lag with pf.
- Mixing peak and RMS values.

Exam tips

- Draw the phasor diagram first.
- ELI the ICE man: V leads I in an inductor.
- pf = cos φ = R/|Z|.
- Check $S^2 = P^2 + Q^2$.

Summary

- $X_L = 2\pi fL$ (I lags); $X_C = 1/(2\pi fC)$ (I leads).
- $Z = R + j(X_L - X_C)$; $I = V/|Z|$.
- $P = VI\cos\varphi$; $Q = VI\sin\varphi$; $S = VI$.
- $pf = \cos\varphi = R/|Z|$.
- Resonance when $X_L = X_C$: $f_0 = 1/(2\pi\sqrt{LC})$.



$\varphi = \tan^{-1}(X/R)$; $pf = \cos\varphi$; $S^2 = P^2 + Q^2$

Quick Quiz

Q1. In a pure inductor, current:
(a) leads V by 90° (b) lags V by 90° (c) is in phase (d) leads by 45°

Ans: (b)

Q2. XC at 50 Hz for C = 100 µF is about:
(a) 3.18 Ω (b) 31.8 Ω (c) 318 Ω (d) 0.03 Ω

Ans: (b)

Q3. R = 6 Ω, XL = 8 Ω: |Z| =
(a) 14 Ω (b) 10 Ω (c) 2 Ω (d) 48 Ω

Ans: (b)

Q4. Power factor of the same circuit:
(a) 0.6 (b) 0.8 (c) 1.0 (d) 0.75

Ans: (a)

Q5. At series resonance:
(a) Z is maximum (b) Z = R (c) I is minimum (d) φ = 90°

Ans: (b)

References

1. **Hughes Edward, Electrical Technology**
   10th Ed., Pearson Education, 2010 (Units 1, 2 textbook)

2. **H. Cotton, Electrical Technology**
   7th Ed., CBS Publishers, 2005 (reference)

Next lecture: Unit 2: single-phase transformers